Elevation Estimation-Driven Building 3-D Reconstruction From Single-View Remote Sensing Imagery

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***Abstract*— Building 3-D reconstruction from remote sensing images has a wide range of applications in smart cities, photogrammetry, and other fields. Methods for automatic 3-D urban building modeling typically employ multiview images as input to algorithms to recover point clouds and 3-D models of buildings. However, such models rely heavily on multiview images of buildings, which are time-intensive and limit the applicability and practicality of the models. To solve these issues, we focus on designing an efficient DSM estimation-driven reconstruction framework (Building3-D), which aims to reconstruct 3-D building models from the input single-view remote sensing image. Existing DSM estimation networks suffer from the imbalance between local and global features, which leads to oversmooth DSM estimates at instance boundaries. To address this issue, we propose a Semantic Flow Field-guided DSM Estimation (SFFDE) network, which utilizes the proposed concept of elevation semantic flow (ESF) to achieve the registration of local and global features. First, in order to make the network semantics globally aware, we propose an elevation semantic globalization (ESG) module to realize the semantic globalization of instances. Furthermore, in order to alleviate the semantic span of global features and original local features, we propose a local-to-global elevation semantic registration (L2G-ESR) module based on ESF. Our Building3-D is rooted in the SFFDE network for building elevation prediction, synchronized with a building extraction network for building masks, and then sequentially performs point cloud reconstruction and surface reconstruction (or CityGML model reconstruction). On this basis, our Building3-D can optionally generate CityGML models or surface mesh models of the buildings. Extensive experiments on ISPRS Vaihingen and DFC2019 datasets on the DSM estimation task show that our SFFDE significantly improves upon state-of-the-art, and δ_1, δ_2, and δ_3 metrics of our SFFDE are improved to 0.595, 0.897, and 0.970. Furthermore, our Building3-D achieves impressive results in the 3-D point cloud and 3-D model reconstruction process.**

***Index Terms*— 3-D building reconstruction, DSM estimation, elevation semantic flow (ESF), remote sensing images.**

I. Introduction

THANKS to the increasing development of various sensors, various remote sensing data have been applied in different fields [1], [2], [3], [4], including semantic segmentation [5], [6], [7] and object detection [8], [9], [10] of 2-D remote sensing image, and 3-D point cloud classification [11], [12], [13]. Among them, the interpretation of interactive information between 2-D data and 3-D data gradually enters researchers' field of vision. 3-D reconstruction of urban buildings is a significant constituent of remote sensing image interpretation, where the goal is to parse image or point cloud data to generate 3-D model representations [14], [15], [16], [17], [18], [19] of urban buildings. In recent years, large-scale geospatial mesh models of urban buildings have been introduced widespread in many fields, such as navigation and urban planning [20] and urban 3-D maps [21].

Among the existing 3-D model acquisition methods, there are mainly four approaches: airborne LiDARs, a reconstruction based on multiview UAV images, and reconstruction based on single-view remote sensing images. Although airborne LiDAR can obtain accurate 3-D point clouds [22], [23], [24], [25], [26], [27] of buildings in the target area, it is limited by high acquisition cost and low acquisition efficiency, and is difficult to apply to large-scale tasks. Building reconstruction based on multiview UAV images [14], [16] can acquire 2-D image data for 3-D model reconstruction at a low cost, but it has

Manuscript received 25 July 2022; revised 6 December 2022, 27 January 2023, 11 February 2023, and 10 March 2023; accepted 4 April 2023. Date of publication 17 April 2023; date of current version 12 May 2023. This work was supported in part by the National Key Technology Research and Development Program of the Ministry of Science and Technology of China under Grant 2018YFE0204203 and in part by the National Key Research and Development Program of China under Grant 2021YFB3900504. *(Corresponding author: Kaiqiang Chen.)*

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Digital Object Identifier 10.1109/TGRS.2023.3266477

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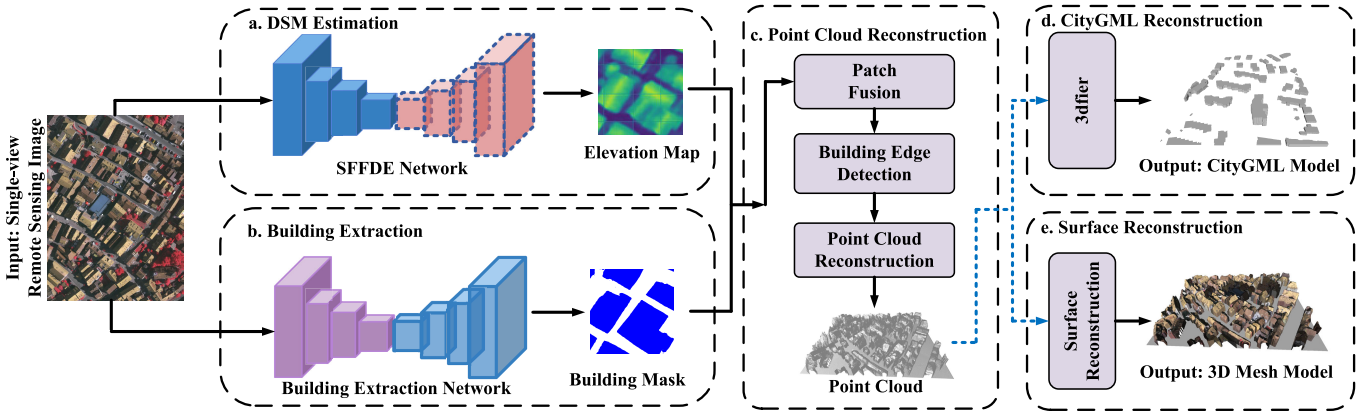


Fig. 1. Flowchart of the proposed Building3-D framework for 3-D reconstruction of buildings. The input of Building3-D is a single-view remote sensing image, and the outputs are the corresponding elevation maps, building masks, point clouds, and 3-D building models. The blue dotted line represents optional follow-up operations (CityGML or mesh models).

a very high time cost for large-scale regional reconstruction and requires a high degree of overlap between images. Unlike airborne LiDAR and drone images, single-view remote sensing imagery can cover a larger area with high stability and low cost. Therefore, we focus on the 3-D reconstruction of buildings based on single-view remote sensing imagery (see Fig. 1), aiming to utilize low-cost large-scale remote sensing imagery for fast urban building reconstruction.

For LiDAR-based methods, the researchers [28] review the existing research progress of building reconstruction methods based on LiDAR (or photogrammetry) point cloud data. Among them, Wang et al. [29] proposed a new method for 3-D building surface modeling based on closed constraints. First, the study utilizes a region-growing algorithm to fit the input point cloud through a set of candidate planes. Then, additional candidate planes are generated from the initial plane according to the rigid transformation, and then, the original set of primitives is extended to the set of candidate models by a generation rule. Li and Wu [30] developed a method for embedding multiple relationships into the procedural modeling process for automatic 3-D reconstruction of buildings from photogrammetric point clouds. It first constructs a hybrid tree of constructive solid geometry and boundary representation (CSG-BRep) and selects cells that approximate the shape of a building according to topological relationship constraints. However, this type of method uses point clouds acquired by expensive LiDARs (or photogrammetry point clouds) for reconstruction, which is expensive and cannot directly use cheap satellite images for building reconstruction. For multiview image-based methods, Hong et al. [31] apply the CasMVSNet-based MVS model to reconstruct 3-D models of postearthquake buildings using multiview drone images to assist in building damage assessment tasks. This method relies on limited multiview images, which requires high image overlap and fine path planning, and cannot directly perform fast building reconstruction with single-view satellite imagery as input.

For single-view-based building 3-D reconstruction methods, some methods [32] derive 3-D building models (e.g., CityGML) and only focus on the restoration of roof structure topology. The researchers aim to reconstruct the roof topology using traditional methods and then obtain 3-D data in the form of CityGML based on the building height information. Some researchers use prelabeled roof structure types to train the network and then extract the roof topology. However, these methods are limited to the extraction of topology structure, which brings unnecessary extraction of artificial prior knowledge for building reconstruction, resulting in a cumbersome and redundant reconstruction process, which is not conducive to building reconstruction in large-scale areas. In order to solve this problem, based on the prediction of building elevation information, we perform 3-D mapping of building elevation to obtain building point cloud data and then selectively reconstruct the subsequent CityGML model or mesh model of the buildings. Based on this, our reconstruction framework (see Fig. 1) can simultaneously reconstruct more realistic mesh models of buildings and virtual models, such as CityGML.

For single-view remote sensing images, the elevation information of buildings is a key factor in constructing 3-D building models. Although substantial progress has been made in existing elevation prediction methods [33], [34], [35], [36], [37], [38] from remote sensing images, the problem of low prediction accuracy occurs when both regular and irregular textured objects are present. Through our investigation, we realize that the low prediction accuracy of textured regular and irregular objects is caused by the inability of the receptive field to achieve a local–global tradeoff. The imbalance between global and local features will cause the network to suffer from the insufficient receptive field in the learning process, and it cannot take into account the extraction of regular texture and irregular texture object features at the same time. Since the features extracted by CNN have a local perception, to solve this problem, we introduce elevation semantic globalization (ESG) to obtain global feature perception. However, the network at this time does not have the ability to take into account and trade off local and global features at the same time. To address this issue, we introduce the concept of elevation semantic flow (ESF) embedded in a local-to-global feature registration

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module to explicitly model the changing trend between global and local features to trade off local and global features. Based on this, we propose an Elevation Semantic Flow Field-guided DSM Estimation (SFFDE) network (see Fig. 1) to obtain accurate elevation predictions for various textured objects.

In this article, we propose a single-view remote sensing image-based building reconstruction framework (Building3-D) based on the SFFDE network, as illustrated in Fig. 1. Unlike previous research schemes targeting roof topology, our Building3-D aims to recover both the CityGML model and the real mesh model of the buildings. Our Building3-D framework consists of four parts: the SFFDE network, the building extraction network, the point cloud reconstruction process, and the building reconstruction process. Given a single-view remote sensing image, the elevation information of the image is first predicted through our SFFDE network, which is mainly composed of two parts: ESG and local-to-global elevation semantic registration (L2G-ESR), which realize the purpose of feature globalization and local and global feature registration, respectively. At the same time, the building mask is extracted from the image through the building extraction network. Next, the building elevation is extracted using the building mask, and the elevations of other uninteresting objects are filtered out. Based on this, the building elevation is used to reconstruct the point cloud data to obtain the 3-D point cloud of the building. Finally, CityGML reconstruction or surface reconstruction is performed using the point cloud to obtain the final CityGML or 3-D mesh model.

The contributions of this study are summarized as follows.

1) We introduce an SFFDE network based on the proposed ESF field for the local-to-global registration of elevation semantics. Specifically, ESG and L2G-ESR are designed to achieve the globalization and local-to-global registration of features, respectively.
2) We propose a single-view image-based building 3-D reconstruction framework (Building3-D) rooted in the proposed SFFDE network and extract building masks according to the building extraction network for subsequent 3-D point cloud and building reconstruction.
3) Experiments on the ISPRS Vaihingen [39] and DFC2019 [27] datasets show that our SFFDE network significantly outperforms the baseline model, achieving new state-of-the-art. What is more, our Building3-D achieves impressive results in the 3-D point cloud and building reconstruction process.

The remainder of this article is organized as follows. In Section II, we give a brief review of the related works on DSM estimation, building extraction, and building reconstruction. The details of our proposed framework Building3-D and SFFDE network are given in Section III. In Section IV, extensive experiments are conducted on the ISPRS Vaihingen dataset and the DFC2019 dataset to demonstrate the effectiveness of our SFFDE network. Also, the 3-D reconstruction experiments of buildings are given in Section IV. Finally, Section V concludes this article.

II. Related Work

A. DSM Estimation

The existing methods [33], [34], [35], [36] for DSM elevation estimation in the remote sensing field are mainly divided into three types: random field-based methods, CNN-based methods, and some hybrid methods. In random field-based methods, researchers [37], [38] utilize conditional random field (CRF) and Markov random field (MRF) to model the local and global structures of an image. Considering that local features cannot provide sufficient features for predicting depth values, Batra and Saxena [37] model the relationship between adjacent regions. Furthermore, in order to obtain global features beyond the local ones, Saxena et al. [38], [40] first compute the features of the four nearest neighbors of the specified patch and then use the MRF and Laplacian models to estimate the depth of each patch. In recent years, CNN has been widely used in the fields of object classification, semantic segmentation, and object detection. Inspired by this, some researchers [41] propose an ResNet-based convolutional network to predict DSM elevation information. In IMG2DSM [42], an adversarial loss function is introduced to improve the possibility of synthesizing DSM. It also employs a conditional generative adversarial network to build image-to-DSM elevation translation. Zhang and Chen [43] improve the learning ability of abstract features of objects of different scales by means of multiscale feature extraction through a multipath fusion network. Li et al. [44] divided the height values into intervals with increasing spacing and transformed the regression problem into an ordinal regression problem, using an ordinal loss for network training. After that, a postprocessing technique is designed to convert the predicted height map of each patch into a seamless height map. Carvalho et al. [45] have studied various loss functions of depth regression in depth, combined the encoder–decoder architecture with adversarial loss, and then proposed a new depth estimation network D3-Net. In the hybrid methods, Wang et al. [46] propose a joint framework incorporating hierarchical CRFs, aiming to predict depth and segmentation outcomes from a single-view remote sensing image. Srivastava et al. [47] use a method for supervised network training via a multitask loss and introduce a unified framework for elevation estimation and semantic segmentation of single-view remote sensing images.

However, none of these methods take into account the need for accurate elevation prediction to simultaneously guarantee the extraction of representative features for both regular and irregular textured objects, which is caused by the imbalance between global and local features. To address this issue, we propose an SFFDE network to trade off local and global features.

B. Building Extraction

Many researchers [48], [49], [50] have devoted themselves to designing efficient methods for automatically extracting buildings, which mainly consists of methods based on handcrafted features and methods based on deep learning. In the handcrafted feature-based methods, geometric, spectral, and

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contextual information of buildings is used to design representative features for accurate building extraction. Lin and Nevatia [48] first apply edge detection to building extraction for the detection of roofs, walls, and shadows. Later, the fractal network algorithm is proposed by Baatz [49] to segment the image at multiple scales and extract the target building by combining the texture and other features of the image. Wang and Liu [50] achieve the purpose of pixel-by-pixel classification of images by using machine learning methods to receive input from feature vectors constructed from image texture, shape, and structural features. However, methods based on handcrafted features often only utilize the shallow features of objects, ignoring the semantics of deep features. Therefore, methods based on deep learning gradually enter the field of vision of researchers. Minh [51] applies an image patch-based building block extraction method to building extraction, an early work on convolutional neural networks applied to building extraction. On this basis, Minh also uses CRF or postprocessing to refine the results of the network. Later, Huang et al. [52] introduce an improved DeconvNet that adds upsampling and cheat connection operations to deconvolution layers to achieve building extraction. Wu et al. [53] propose a multiconstraint FCN to perform feature learning on remote sensing images for the purpose of automatically extracting buildings.

In this article, we apply the popular segmentation network DeeplabV3+ [54] to generate binary masks for buildings.

C. Building Reconstruction

The existing methods for 3-D building reconstruction mainly focus on roof topology restoration. Researchers [55], [56], [57], [58] extract roof vertices, eave lines, or segmented planes as primitives from an image or DSM through image segmentation, edge detection, plane patch detection, and so on. After that, the combination, segmentation, and topological analysis of the individual geometric primitives are executed in sequence to generate the 3-D shapes of the buildings. Bulatov et al. [55] extract building regions by performing vegetation and outlier filtering on a normalized DSM (nDSM) and then combine the extracted building pixels with operations such as graph-based orthophoto segmentation, dominant direction extraction, polygonization, and generalization to extract buildings contour. Finally, they construct the building model by topologically analyzing the roof details. Li et al. [56] classify MVS point clouds through graph-cut-based MRFs, followed by RANSAC and regularized MRF for optimization and roof structure extraction, respectively.

However, these methods suffer from the limitations of multiview image input or surfel geometric topology, thus bringing great difficulties to building reconstruction in large scenes. To address this issue, we propose a DSM estimation-driven framework for building 3-D reconstruction with single-view remote sensing images as input.

III. Our Approach

A. Overview

The flowchart of our proposed building 3-D reconstruction framework (Building3-D) is illustrated in Fig. 1. As a single-view image-based building 3-D reconstruction approach, our framework consists of four stages: the DSM estimation branch, the building extraction branch, the point cloud reconstruction branch, and the building reconstruction branch (surface reconstruction or CityGML reconstruction). Given an input remote sensing image, the DSM estimation branch and the building extraction branch first come into play to estimate the elevation information and extract the buildings, respectively. On this basis, the information obtained by the building extraction branch is used to filter the information from the elevation results predicted in the DSM estimation branch to obtain only the elevation information of buildings. Next, the point cloud data corresponding to the building are recovered according to the input image and the predicted elevation information. Finally, surface reconstruction or CityGML reconstruction is performed on the point cloud of buildings obtained by the point cloud reconstruction branch to obtain the final mesh or CityGML model. In the following, we first introduce the SFFDE network and then introduce the framework's building extraction process, point cloud reconstruction process, and building reconstruction process.

B. Semantic Flow Field-Guided DSM Estimation Network

Considering generating fine 3-D point clouds, accurate DSM information is necessary. To address this issue, we design an SFFDE network, as shown in Fig. 2. Our SFFDE selects PSPNet [59] as the baseline and consists of three stages. First, a single remote sensing image is sent into a feature extraction network (such as resnet101) and a pyramid pooling module (PPM) [59] module to obtain high-level representations of features. Second, an ESG module is introduced for the globalization of semantic features to obtain high-level global semantic representation. Finally, we propose an L2G-ESR module to achieve the registration of low-level local semantics and high-level global semantics of features between different resolutions. Furthermore, our SFFDE network is driven by the berHuLoss [60] during training.

1) Elevation Semantic Globalization: The stacking of convolutional layers and pooling layers can increase the receptive field, but the receptive field of the convolution kernel of a specific layer on the original image is limited, which is unavoidable in local operations. However, elevation estimation based on a single-view remote sensing image requires more information on the original image. If the global information can be introduced in some layers, the problem that the local operation of the convolution cannot capture the global information can be well solved, and it can bring richer information to the subsequent layers. Inspired by the transformer [61] architecture, we introduce the ESG module (see Fig. 2) for the global semantic representation.

Given feature maps $\mathbf{F} \in \mathbb{R}^{H\times W\times C}$ (H, W, and C stand for height, width, and the number of channels, respectively), after the feature extraction of the backbone, these feature maps

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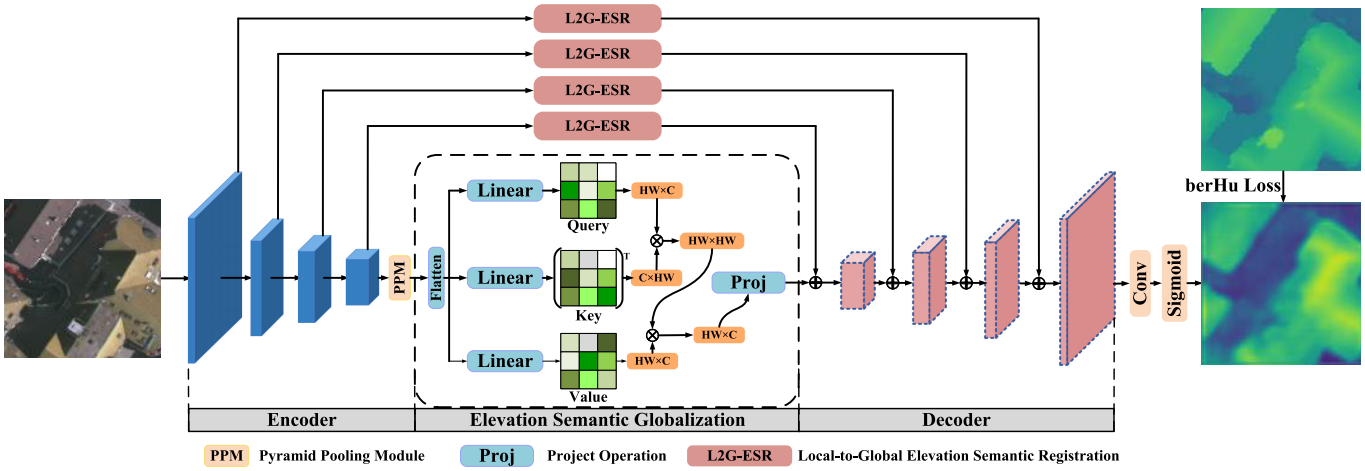


Fig. 2. Flowchart of the proposed SFFDE network for DSM estimation. The proposed ESG is responsible for the global semantic representation. Furthermore, L2G-ESR is introduced to achieve the tradeoff of local and global features.

are high-level semantic features with local perception. Before being sent to the ESG module, the feature map $\mathbf{F}$ is stretched from a 2-D raster image into a 1-D vector (regardless of the channel dimension), which can be formulated as

$$\mathcal{F} = \text{flatten}(\mathbf{F}) \tag{1}$$

where $\mathcal{F} \in \mathbb{R}^{N\times C}$ $(N = H \times W)$. The stretched feature $\mathcal{F}$ goes through three fully connected layers in parallel for feature transformation so that the 2-D raster image features are mapped to the same space as the 1-D vector features, which is formulated as

$$\begin{aligned}\left\{\mathbf{Q}, \mathbf{K}, \mathbf{V}\right\} &= \left\{\text{FC}_1(\mathcal{F}), \text{FC}_2(\mathcal{F}), \text{FC}_3(\mathcal{F})\right\} \\ &= \left\{\mathbf{W}_q\mathcal{F}, \mathbf{W}_k\mathcal{F}, \mathbf{W}_v\mathcal{F}\right\}\end{aligned} \tag{2}$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N\times D}$. D is the number of channels of the output features. Given query embeddings **Q**uery to be enhanced and feature maps **K**ey, **V**alue to be fused, the ESG operation is defined as

$$\mathbf{F}_{\text{esg}} = \text{ESG}\left(\mathbf{Q}, \mathbf{K}, \mathbf{V}\right) = \text{Softmax}\left(\mathbf{W}_q\mathcal{F}(\mathbf{W}_k\mathcal{F})^\top\right)\mathbf{W}_v\mathcal{F} \tag{3}$$

where $\mathbf{F}_{\text{esg}} \in \mathbb{R}^{N\times D}$ is the query feature after ESG operation.

After this, we set up a multihead ESG operation module based on ESG operation and transformer architecture; the formula is given as follows:

$$\mathbf{F}_{\text{out}} = \text{MHE}(\mathbf{W}_q\mathcal{F}, \mathbf{W}_k\mathcal{F}, \mathbf{W}_v\mathcal{F}) \tag{4}$$

where $\mathbf{F}_{\text{out}}$ is the output features of MHE. MHE is the multihead ESG operation that is similar to the multihead attention (MHA) operation [61]. Specifically, each ESG head can be expressed as

$$\text{Head}_i = \text{ESG}(Q, K, V) \tag{5}$$

where $Q, K,$ and V represent the **Q**uery, **K**ey, and **V**alue matrix, respectively. Then, based on this, multihead ESG can be expressed mathematically as

$$\text{MultiHead}(Q, K, V) = \text{Cat}(\text{Head}_1, \ldots, \text{Head}_8)W \tag{6}$$

where Cat is the concatenate operation and W represents the weight of the final full connection operation. Different ESG heads represent different subspaces, and the results of all ESG heads are spliced together to obtain the final result through full connection.

Next, a feature projection operation is adopted to perform feature mapping between spatial 1-D features and spatial 2-D features. Formally, the obtained projection feature $\mathbf{F}_{\text{proj}}$ can be defined as

$$\mathbf{F}_{\text{proj}} = \text{Relu}\Big(\text{LN}\big(\text{FC}(\mathbf{F}_{\text{out}})\big)\Big) \tag{7}$$

where LN is the LayerNorm [62] operation. Finally, the projection feature $\mathbf{F}_{\text{proj}} \in \mathbb{R}^{N\times D}$ is reshaped to the same resolution as $\mathbf{F}$.

2) Elevation Semantic Flow: To register features at different resolutions, inspired by Optical Flow, we introduce the concept of ESF. The Optical Flow is the method that employs the changes of pixels in the image sequence in the time domain and the correlation between adjacent frames to find the corresponding relationship between the previous frame and the current frame, so as to calculate the motion of objects between adjacent frames. Analogous to the instantaneous gray change rate of pixels at the same location between video frames in a short period of time, we define a semantic change rate of elevation, that is, the semantic displacement of pixels between adjacent resolution images, which can be expressed as

$$\vec{u} = \left(\frac{\partial x}{\partial l}, \frac{\partial y}{\partial l}\right) \tag{8}$$

where l represents the varying resolution of feature maps and $\vec{u}$ is the ESF vector, which represents the rate of change of semantics along the x- and y-directions.

In space, motion can be described by a motion field. In the video, the motion of an object is often represented by an optical flow field. In the feature map of the elevation estimation network, the semantic motion can be defined by the feature vectors represented by the pixels of the feature maps of different resolutions. Thus, we propose to define an elevation

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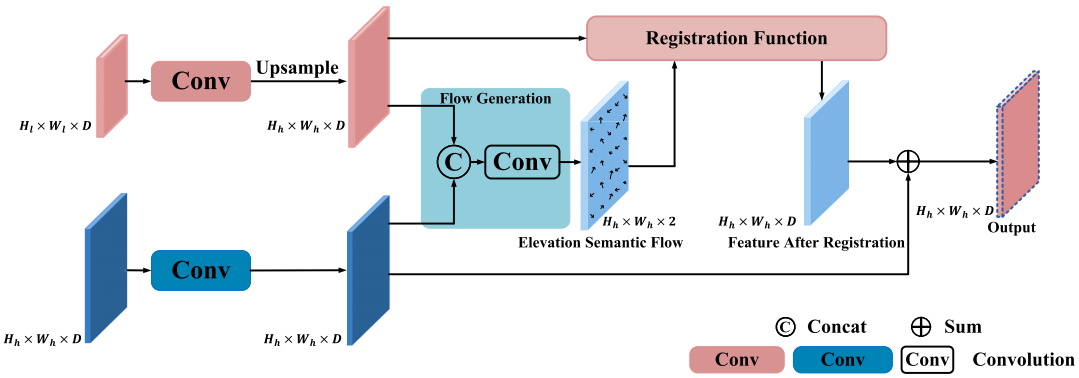


Fig. 3. Overview of our L2G-ESR module. The inputs to L2G-ESR are two features of different resolutions. Flow generation and feature registration are performed sequentially. The pink and blue convolution blocks in the figure represent feature mapping operations for low resolution and high resolution, respectively. Specifically, both convolution operations are implemented using 1×1 2-D convolution, and the input and output channel dimensions are the same.

semantic field of F, which is formulated as

$$-\frac{\partial F}{\partial l} = \nabla F \cdot \vec{u} = \frac{\partial F}{\partial x}\frac{\partial x}{\partial l} + \frac{\partial F}{\partial y}\frac{\partial y}{\partial l}. \tag{9}$$

Similar to the optical flow field, the elevation semantic field is a 2-D vector field, which reflects the semantic change trend of each point in the feature map. Classical optical flow estimation is solved by a linear algebra method (such as the LK algorithm [63]). We use an end-to-end training nonlinear optimization method and use L2G-ESR as a constraint to learn the ESF field.

3) Local-to-Global Elevation Semantic Registration: After the globalization of features, the perceptual ability of features is extended from local to global. This enables the network to not only perceive low-dimensional local features but also acquire the ability to perceive high-dimensional global features. However, since low-dimensional local features and global high-dimensional features are obtained by local convolution and ESG (not convolution operations), respectively, there is a huge span in the range of feature perception between global features and local features. In addition, these features do not have both local and global perception capabilities at the same time. To address these issues, L2G-ESR is introduced, as shown in Fig. 3.

Given the high-resolution local feature $\mathbf{F}_h$ and the low-resolution globalized feature $\mathbf{F}_l$, we first map $\mathbf{F}_h$ and $\mathbf{F}_l$ to the same number of channels as

$$\left\{\widehat{\mathbf{F}}_h, \widehat{\mathbf{F}}_l\right\} = \left\{\text{Conv}_1(\mathbf{F}_h), \text{Conv}_2(\mathbf{F}_l)\right\} \tag{10}$$

where $\widehat{\mathbf{F}}_h \in \mathbb{R}^{H_h \times W_h \times D}$ and $\widehat{\mathbf{F}}_l \in \mathbb{R}^{H_l \times W_l \times D}$ are the new features after mapping. Conv_1 and Conv_2 correspond to the blue and pink convolution blocks in the figure, respectively. Specifically, both convolution operations are implemented using 1×1 2-D convolution, and the input and output channel dimensions are the same. Then, we bilinearly interpolate the new low-resolution features $\widehat{\mathbf{F}}_l$ to the same resolution as the high-resolution features $\widehat{\mathbf{F}}_h$. On this basis, the two same-resolution features are fused and further encoded into a 2-D ESF field. Mathematically, the ESF field $\mathbf{S}$ can be expressed as

$$\mathbf{S} = \text{Conv}\Big(\text{cat}\big(\text{upsample}(\widehat{\mathbf{F}}_l), \widehat{\mathbf{F}}_h\big)\Big) \tag{11}$$

where upsample represents the bilinear interpolation operation, cat is the concatenation operation, and Conv denotes the convolution operation. The semantic flow field $\mathbf{S} \in \mathbb{R}^{H_h \times W_h \times 2}$ represents the changing trend of elevation semantics between different resolutions. The channel "2" refers to the change of semantics in both x- and y-directions.

Let $\mathbf{L} \in \mathbb{R}^{H_h \times H_h \times 2}$ denote the coordinates of each pixel in the feature map $\widehat{\mathbf{F}}_h$. Then, the generated semantic flow field $\mathbf{S}$ is employed to get the coordinates $\widehat{\mathbf{L}}$ of each pixel of the feature map after offset, as $\widehat{\mathbf{L}} = \mathbf{L} + \mathbf{S}(x' = x + \Delta x, y' = y + \Delta y)$.

Following the bilinear sampling method in STN [64], the pixel $\mathbf{F}_{\text{reg};ab}$ at (a, b) of the output feature $\mathbf{F}_{\text{reg}}$ after feature registration operation is defined as

$$\begin{aligned}\mathbf{F}_{\text{reg};ab} &= R(\widehat{\mathbf{F}}_l) \\ &= \sum_{m=1}^{H}\sum_{n=1}^{W} \text{upsample}(\widehat{\mathbf{F}}_l)_{mn} \cdot \max(0, 1 - |\widehat{\mathbf{L}}_{x;ab} - m|) \\ &\quad \cdot \max(0, 1 - |\widehat{\mathbf{L}}_{y;ab} - n|)\end{aligned} \tag{12}$$

where $\text{upsample}(\widehat{\mathbf{F}}_l)_{mn}$ is the pixel at (m, n) of the aforementioned $\widehat{\mathbf{F}}_l$ after upsampling. Furthermore, $\widehat{\mathbf{L}}_{x;ab}$ and $\widehat{\mathbf{L}}_{y;ab}$ are the x and y coordinates of each pixel of $\widehat{\mathbf{L}}$. After that, the feature $\mathbf{F}_{\text{reg}}$ after registration is fused with aforementioned $\widehat{\mathbf{F}}_h$, as

$$\mathbf{F}_{\text{out}} = \mathcal{I}(\mathbf{F}_{\text{reg}}, \widehat{\mathbf{F}}_h) \tag{13}$$

where $\mathcal{I}$ is the aggregation function between the feature $\mathbf{F}_{\text{reg}}$ after registration and the feature $\widehat{\mathbf{F}}_h$.

4) Loss Function: In order to enable the network to better regress the elevation value of the instance, we use berHuloss [60] as our loss function, which is expressed as

$$\mathcal{L}(x) = \begin{cases} |x|, & |x| \le c \\ \dfrac{x^2 + c^2}{2c}, & |x| > c \end{cases} \tag{14}$$

where c is the judgment threshold. Specifically, $c = 0.2 \times \max(|\text{predict} - \text{gt}|)$ in our experiments.

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C. Building Extraction Network

Unlike regular segmentation networks, we only focus on the single category (buildings) of elements. In the building mask extraction process, we first perform binarization preprocessing on the labels of the dataset. Specifically, the label value of the pixel where the building is located is set to 1, and the label value of the nonbuilding pixel is set to 0. In order to better extract building masks, we use the currently popular Deeplabv3+ [54] as our building extraction network to extract the building mask $\mathbf{M}$, which is built on top of the backbone ResNet-101. Given an input image, the building extraction network outputs a mask $\mathbf{M}$ of the building. Specifically, the building pixel value is 1, and the background is 0. During training, the update of network parameters is driven by the cross-entropy loss function.

What is more, the canny edge detection algorithm is used to extract the outlines of buildings. The extracted outlines and building masks with fine pixel positioning give the next procedures the precise building positioning.

D. Building Reconstruction

Building 3-D reconstruction based on remote sensing imagery has always been a hot research topic. However, most of the existing methods are limited to a single output of the building CityGML model, and a real 3-D model of buildings has not been constructed. To address this issue, we propose a building 3-D reconstruction method based on building elevation information predicted by our SFFDE (see Fig. 1). After the aforementioned steps, we adopt the building masks $\mathbf{M}$ to extract the elevation information of the buildings and filter out the categories (such as vegetation) that are not of interest, which is formulated as

$$\mathbf{E}_{\text{building}} = \mathbf{E} \circ \mathbf{M} \tag{15}$$

where $\mathbf{E}$ and $\mathbf{E}_{\text{building}}$ are the elevation information before and after filtering. $\circ$ is the Hadamard product operation. After obtaining the elevation information of the buildings, the two procedures of point cloud reconstruction and building reconstruction (surface reconstruction or CityGML reconstruction) are performed sequentially.

1) Point Cloud Reconstruction: Generating the point cloud data of the object is a key step to obtain the 3-D structure model of the object. In this process, the obtained DSM prediction results of each input image patch are first fused through patch fusion to obtain a larger area of the scene. After that, in order to smooth the gap between patches, the Gaussian filtering operation is employed to the DSM predicted results of each large scene remote sensing image. Then, we perform 2-D-to-3-D mapping of buildings based on the elevation information of large-area building clusters to generate their 3-D point cloud data. The latitude and longitude coordinates corresponding to the pixels are used as the x, y coordinates of the point cloud, and the elevation information is used as the z coordinates of the point cloud.

2) Mesh and CityGML Model: Mesh refers to a polygonal grid, which is a data structure used in computer graphics for modeling various irregular objects. In the face of the polygon mesh, the triangular face is the smallest unit to be divided, so it is often referred to as the triangular face. The basic components of a mesh are vertices, edges, and faces. CityGML is a data format used to construct virtual 3-D city models and is a general data model used to express 3-D city templates. CityGML can not only express the graphic appearance of the city model but also take care of the semantic representation, such as the classification and aggregation of digital terrain models, and vegetation and water systems. All models can be divided into five different coherent levels of detail (LOD), with increasing LOD to obtain more details about the geometry and themes. The five consecutive LODs are LOD0, LOD1, LOD2, LOD3, and LOD4. The CityGML model reconstructed in this article is the LOD1 model in five coherent LODs.

3) Surface Reconstruction: Based on the point cloud data of the building generated in the previous steps, we first normalize the point cloud to facilitate the subsequent reconstruction process. Then, we perform Poisson [65] reconstruction on the normalized point cloud data to obtain the mesh model of the building.

4) CityGML Reconstruction: To verify the effectiveness of our method, we also extend the SFFDE network to the algorithm for building CityGML model reconstruction in our Building3-D. After our Building3-D performs the 3-D point cloud mapping operation, we also extract the polygonal structure of the building from the image at the same time. Combining the obtained 3-D point cloud and building polygon structure, we use 3dfier [71] to reconstruct the building CityGML model from single-view remote sensing images.

IV. Experiments

In this section, we first give an introduction to the datasets used by our entire framework. Next, we introduce the specific experimental setup (including evaluation protocols and completion details) in the experiments. Then, we show the performance and visualization results of our proposed SFFDE on elevation estimation. After that, a reconstruction analysis of the entire reconstruction framework Building3-D is presented. Finally, to demonstrate the effectiveness of the proposed ESG operation and the L2G-ESR operation, we conduct extensive ablation experiments.

A. Datasets

1) ISPRS Vaihingen: There are a total of 33 slices in the ISPRS Vaihingen [39] dataset, and each slice has approximately 2500 $\times$ 2500 pixels. Each remote sensing image is accompanied by orthophoto images, semantic labels, and digital surface models (DSM and nDSM). The ground sampling distance (GSD) of each image is 9 cm, and it has three channels of near-infrared, red, and green. According to the official split, 16 slices that provided ground truth are used for the training of models, and the remaining 17 slices are used for evaluation by the challenger organizer.

2) 2019 Data Fusion Contest: The 2019 Data Fusion Contest (DFC2019) [27] dataset currently includes approximately 100 km^2 of coverage for Jacksonville, FL, USA, and Omaha, NE, USA. The GSD is about 30 cm, and each image with

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TABLE I

PERFORMANCE OF DSM ESTIMATION ON THE ISPRS VAIHINGEN DATASET. ↑ MEANS THAT THE HIGHER THE INDICATOR VALUE, THE BETTER THE PERFORMANCE, AND ↓ MEANS THAT THE LOWER THE INDICATOR VALUE, THE BETTER THE PERFORMANCE

Method	Rel↓	RMSE↓/m	RMSE(log)↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
Amirkolaee *et al.* [41]	1.163	2.871	0.334	0.330	0.572	0.741
IMG2DSM [42]	-	2.58±0.09	-	-	-	-
D3Net [45]	2.016	2.123	-	0.369	0.533	0.644
Li *et al.* [44]	0.314	1.698	0.155	0.451	0.817	0.939
PSDNet [66]	0.363	1.542	0.171	0.447	0.745	0.906
ASSEH + ResNet50 [67]	0.272	1.380	0.146	0.468	0.788	0.951
ASSEH + ResNet101 [67]	0.237	1.144	0.120	0.595	0.860	**0.971**
Adabins [68]	0.286	1.536	-	0.513	0.808	0.937
LeReS [69]	0.260	1.612	-	0.554	0.800	0.932
WMD [70]	0.272	1.692	-	0.543	0.798	0.916
SFFDE + ResNet50 (ours)	0.225	1.145	0.087	**0.624**	0.841	0.933
SFFDE + ResNet101 (ours)	**0.222**	**1.133**	**0.084**	0.595	**0.897**	0.970

semantic labels and nDSM is 1024 × 1024 pixels in size. The dataset provides WorldView-3 panchromatic and eight-band visible and near-infrared (VNIR) images. DFC2019 includes 26 images collected in Jacksonville and 43 images collected in Omaha.

B. Experimental Settings

1) Evaluation Protocols: We use six metrics [72] to evaluate the DSM estimation performance of our proposed method, including mean relative error (Rel), RMSE, RMSE(log), and the ratio of pixels with predicted elevation values close to the ground truth (δ_1, δ_2, and δ_3). Among them, mean Rel, RMSE, and RMSE(log) are expressed as

$$\text{Rel} = \frac{1}{N}\sum_{i=1}^{N}\frac{\left|D_i - D_i^*\right|}{D_i^*} \tag{16}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left|D_i - D_i^*\right|^2} \tag{17}$$

$$\text{RMSE(log)} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left|\log D_i - \log D_i^*\right|^2} \tag{18}$$

where N is the number of the pixels, D_i is the predicted elevation value of the ith pixel, and D_i^* is the ground truth of the ith pixel. What is more, δ_i is expressed as

$$\delta_i : \max\left(\frac{h_{\text{pred}}}{h_{\text{gt}}}, \frac{h_{\text{gt}}}{h_{\text{pred}}}\right) < 1.25^i \tag{19}$$

where h_{gt} and h_{pred} are the ground truth and predicted elevation values. In the specific, select the maximum value of $h_{\text{pred}}/h_{\text{gt}}$ and $h_{\text{gt}}/h_{\text{pred}}$, and count the percentage of all pixels in h_{pred} to the overall pixels that are smaller than the threshold 1.25^i. The closer the value of δ_i is to 1, the better the effect.

2) Implementation Details: Our framework is implemented based on the PyTorch Library. The momentum SGD algorithm with the momentum value set to 0.9 is employed to optimize both the DSM estimation branch and the building extraction branch. We train our model for 80 000 iterations, and the initial and minimum learning rates are set to 0.005 and 0.00002, respectively. The weight decay value is set to 0.0005 for regularization. Our DSM estimation branch and building extraction branch are both executed on a single NVIDIA TITAN RTX GPU with batch size set to 4. Considering that the huge size of each image in both Vaihingen and DFC2019 datasets makes the images unable to directly be sent to the network due to the GPU memory limit, we employ a sliding window strategy to generate small image patches with 512 × 512 pixels.

When evaluating the accuracy, for each dataset, we only select the checkpoint saved in the last iteration (80 000 iterations) for testing. The number of training iterations is set in consideration of ensuring that the model is converged and stable. The choice of the checkpoint is based on the weight file saved in the last iteration.

C. Performance Analysis of DSM Estimation

1) ISPRS Vaihingen: We compared the DSM estimation performance by our SFFDE with other prediction networks, including recent methods: PSDNet [66], ASSEH [67], Adabins [68], LeReS [69], and WMD [70]. In Table I, the comparisons between our SFFDE network and these methods on the ISPRS Vaihingen dataset are given. It is clear that our SFFDE network achieves the highest performance for the elevation estimation task. Regardless of choosing ResNet50 or ResNet101 as the backbone, our SFFDE outperforms state-of-the-art methods on all metrics (δ_3: comparable to best performance). Rel, RMSE, and RMSE(log) are indicators that describe the prediction error of the model. Among these metrics, with the ResNet101 backbone, our SFFDE achieves 0.222 Rel, 1.133 RMSE, and 0.084 RMSE(log), achieving the best performance. This shows that the prediction results of our SFFDE maintain a high consistency with the ground truth. What is more, δ_1, δ_2, and δ_3 are indicators describing the prediction accuracy of the model. Note that, for the indicators (δ_1, δ_2, and δ_3), SFFDE achieves the highest δ_1 and δ_2 values, and is comparable to the best performance on δ_3, which strongly demonstrates that the elevation value predicted by our SFFDE is close to the ground-truth elevation value. This

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TABLE II

PERFORMANCE OF DSM ESTIMATION ON THE DFC2019 DATASET. ↑ MEANS THAT THE HIGHER THE INDICATOR VALUE, THE BETTER THE PERFORMANCE, AND ↓ MEANS THAT THE LOWER THE INDICATOR VALUE, THE BETTER THE PERFORMANCE

Method	Rel↓	RMSE(log)↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
D3Net [45]	0.526	0.208	0.256	0.635	0.846
DORN [73]	0.488	0.200	0.317	0.646	0.859
FastDepth [74]	0.383	0.189	0.384	0.701	0.875
PSDNet [66]	0.429	0.192	0.375	0.706	0.891
ASSEH + ResNet50 [67]	0.491	0.206	0.334	0.654	0.863
ASSEH + ResNet101 [67]	0.339	0.173	0.448	0.753	0.905
SFFDE + ResNet50 (ours)	**0.272**	0.029	**0.601**	0.778	0.882
SFFDE + ResNet101 (ours)	0.330	**0.024**	0.492	**0.782**	**0.908**

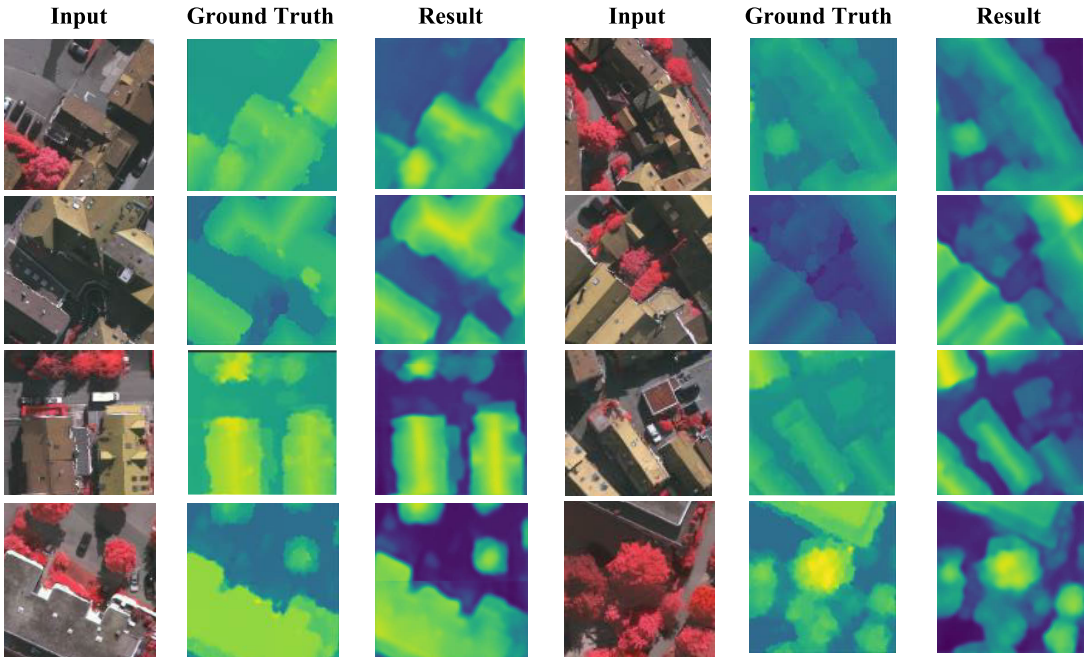


Fig. 4. Visualizations of the predicted elevation results (512 × 512 patches) of our SFFDE network on the ISPRS Vaihingen dataset.

again proves that the local and global semantic registration implemented by SFFDE can improve the accuracy of elevation prediction.

The reason is that our SFFDE can use ESG to achieve global feature extraction and L2G-ESR to achieve global and local feature registration. Based on the concept of ESF, we can well explicitly model the semantic change rate of semantic features describing elevation across different resolutions.

2) DFC2019: In Table II, our SFFDE network is compared with other state-of-the-art DSM estimation methods (including recent method PSDNet [66] and ASSEH [67]) on the DFC2019 dataset. It is clear that our SFFDE outperforms all existing elevation estimation methods on the DFC2019 dataset. Table II displays the evaluation estimation results of our SFFDE network. Whether we choose ResNet50 or ResNet101 as our backbone, SFFDE again outperforms state-of-the-art methods on all indicators. For the ResNet101 backbone, SFFDE improves FastDepth by **0.108** (0.492 versus 0.384) on δ_1, **0.081** (0.782 versus 0.701) on δ_2, and **0.033** (0.908 versus 0.875) on δ_3, and improves ASSEH + ResNet101 by **0.044** (0.492 versus 0.448) on δ_1, **0.029** (0.782 versus 0.753) on δ_2, and **0.003** (0.908 versus 0.905) on δ_3, which are large margins for the challenging DSM estimation problem. Compared with the ISPRS Vaihingen dataset, the DFC2019 dataset has more complex scenes and diverse instances, which undoubtedly brings great difficulties to the DSM estimation problem. However, our SFFDE not only performs the best on the ISPRS Vaihingen dataset but also has the highest performance on the DFC2019 dataset, which once again proves that our proposed feature registration based on the ESF field can well improve complex scenes elevation prediction performance.

D. Visualization Analysis

1) DSM Visualizations on ISPRS Vaihingen: As shown in Fig. 4, local area visualizations of the DSM estimation results of SFFDE are given on the ISPRS Vaihingen dataset. From the visualized results, we can conclude that our SFFDE has high accuracy for building elevation prediction, and the prediction results are basically consistent with the ground truth. Furthermore, the visualization clearly shows that our predicted

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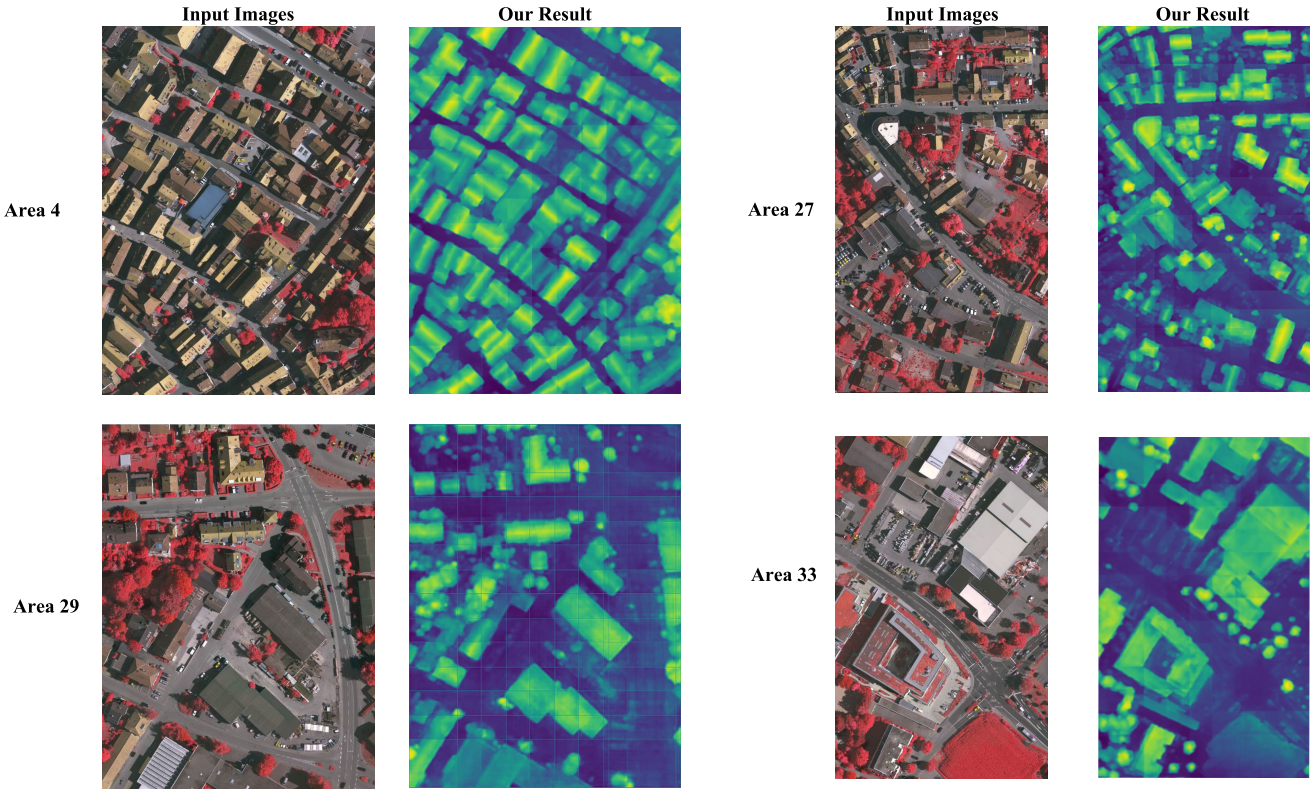


Fig. 5. Visualizations of the predicted elevation results (large areas) of our SFFDE network on the ISPRS Vaihingen dataset.

TABLE III

ABLATION STUDIES ON THE ISPRS VAIHINGEN DATASET. "ESG" DENOTES ELEVATION SEMANTIC GLOBALIZATION. "L2G-ESR" REPRESENTS LOCAL-TO-GLOBAL ELEVATION SEMANTIC REGISTRATION. ↑ MEANS THAT THE HIGHER THE INDICATOR VALUE, THE BETTER THE PERFORMANCE, AND ↓ MEANS THAT THE LOWER THE INDICATOR VALUE, THE BETTER THE PERFORMANCE

Method	Rel↓	RMSE↓/m	RMSE(log)↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
ResNet-50						
Baseline	0.367	1.330	0.135	0.357	0.618	0.819
+ESG	0.277	1.188	0.109	0.542	0.800	0.915
+ESG + L2G-ESR (ours)	**0.225**	**1.145**	**0.087**	**0.624**	**0.841**	**0.933**
ResNet-101						
Baseline	0.358	1.293	0.130	0.374	0.701	0.870
+ESG	0.276	1.282	0.111	0.534	0.843	0.952
+ESG + L2G-ESR (ours)	**0.222**	**1.133**	**0.084**	**0.595**	**0.897**	**0.970**

instance elevations are well-defined. At instance boundaries, our method has small prediction errors and aliasing.

As shown in Fig. 5, we also present the visualization results of the elevation prediction for a large area of the ISPRS Vaihingen dataset. The images of three large scenes (Area 4 contains a large number of buildings, Areas 27 and 29 have both regular and irregular texture instances, and Area 33 has more interior details of buildings) are selected as the inputs of our SFFDE, and the prediction results are visualized. From the visualization results of Area 4, our method is able to make good elevation predictions for areas with dense buildings. In these densely built areas, the buildings are all structures with high roofs in the center and low on both sides. Nonetheless, our visualizations fit this phenomenon well, yielding high predictive performance. This is highlighted in the middle of the roof in the picture, and the sides are darker to demonstrate that Areas 27 and 29 contain a large number of regular and irregular instances, such as buildings and trees. For the regions where these regular and irregular instances coexist, this requires the model to have high generalization performance. From the visualization results, our SFFDE achieves superior elevation prediction performance and can guarantee high accuracy for both regular and irregular texture instances. This benefits from the higher performance of our proposed ESF-based feature registration. Furthermore, our SFFDE can predict the detailed structure of buildings well, as can be seen from the results for Area 33. On top of complex buildings, smooth roofs and complex structures coexist. In our prediction results, the high accuracy of the smooth roof prediction is reflected in the smoothness of the elevation prediction of the roof, which is

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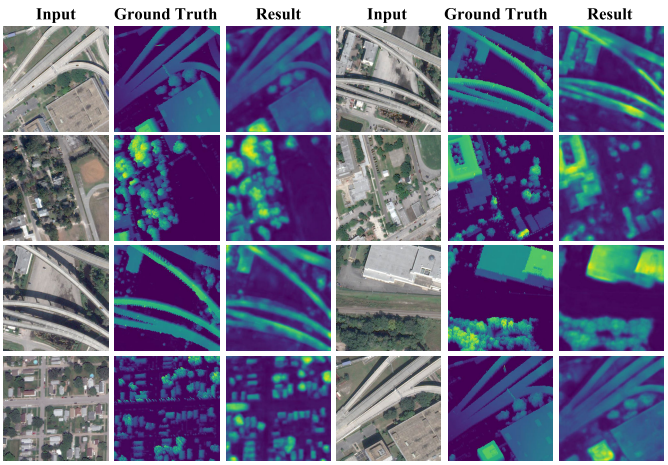


Fig. 6. Visualizations of the predicted elevation results (512 × 512 patches) of our SFFDE network on the DFC2019 dataset.

proven by the basically consistent colors of the smooth areas in the figure. In addition, in the region of complex structure, our SFFDE gives fine boundary structure prediction results. This is superior to other advanced methods.

2) DSM Visualizations on DFC2019: To verify the performance of our SFFDE, we also conduct experiments on the DFC2019 dataset and give high performance. Furthermore, we present the prediction visualization results of our SFFDE on the DFC2019 dataset in Fig. 6. Since DFC2019 contains complex instances, such as bridges, buildings, and trees, our visualization results all contain these complex instances. It can be seen from the visualization results that the elevation information of various instances that we predicted is basically consistent with the ground truth. Furthermore, for regularly textured objects, such as buildings and bridges, our prediction results have clear boundary information and smooth internal structures, which are extremely challenging problems in elevation prediction tasks. Nonetheless, our SFFDE exhibits high prediction performance, which validates the effectiveness of our method. In addition, we also achieved high prediction accuracy for objects with irregular textures, such as trees.

3) Feature Map Visualizations: Feature map visualizations on the ISPRS Vaihingen dataset are given, as shown in Fig. 9. As noted in the figure, the visualization consists of four columns, which are the input image, the feature map before globalization and registration, the feature map after globalization, and the feature map after global–local semantic registration. From the feature map after globalization, we can infer that the feature obtains a global receptive field, not only limited to local perception. In addition, the registered feature map can clearly observe the object and its detailed information, the internal features of the instance are smooth, and the pixel values tend to be continuous. This matches well with better regression problems. Therefore, from the visualization results of the two-part feature maps, it can be seen that our network can simultaneously ensure the feature's ability to perceive the global and the local perception of details and achieve a better balance between global and local features.

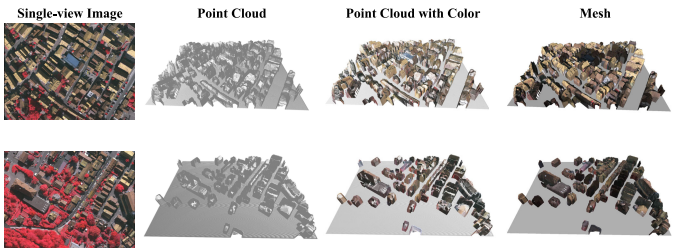


Fig. 7. Visualizations of point cloud and surface mesh of the input single-view image. The first column is the input single-view images, the second column is the reconstructed point clouds, the third column is the reconstructed point clouds with color, and the last column is the reconstructed mesh models of the input images.

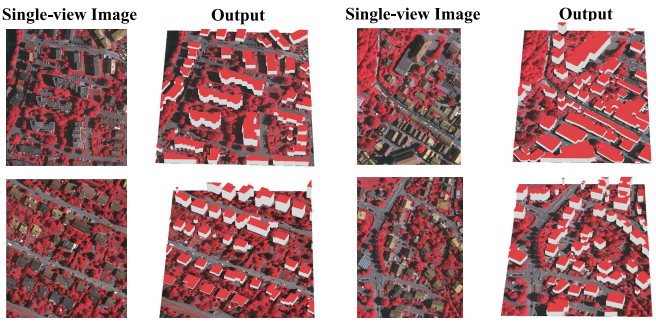


Fig. 8. Visualizations of LOD1 model of buildings in the input single-view image.

E. Building3-D Reconstruction Analysis

1) Building Extraction: As shown in Fig. 10, the visualization results of the building extraction network in the Building3-D framework are given. We not only visualize the local building area in Vaihingen but also visualize the extraction results of building groups in a large area. Based on the superior building extraction network, we can see that the extracted building area is basically consistent with the actual building area, achieving extremely high building extraction performance. This lays a good foundation for subsequent building elevation extraction, building point cloud reconstruction, and surface reconstruction.

2) Surface Reconstruction: Given a single-view remote sensing image of an area, after the building elevation prediction and building extraction steps mentioned above are completed, the next process is to reconstruct the building in 3-D. Based on the extracted building areas, we filter the predicted elevation information to obtain building elevations. The visualization results of building surface reconstruction based on single-view remote sensing images are given in Fig. 7. Given a single-view remote sensing image, our Building3-D outputs a point cloud and mesh of the building. From Fig. 7, we can see that the building has been converted into a 3-D structure, while other elements, such as cars and vegetation, have not been converted. The first column in Fig. 7 shows the input single-view image. On the basis of the obtained building elevations, we perform 3-D mapping on the recovered elevation information to obtain 3-D point clouds. The results of the 3-D point cloud visualization of the mapped buildings are given in the second column of the figure. In order to better show the point cloud results, we also attached the color to

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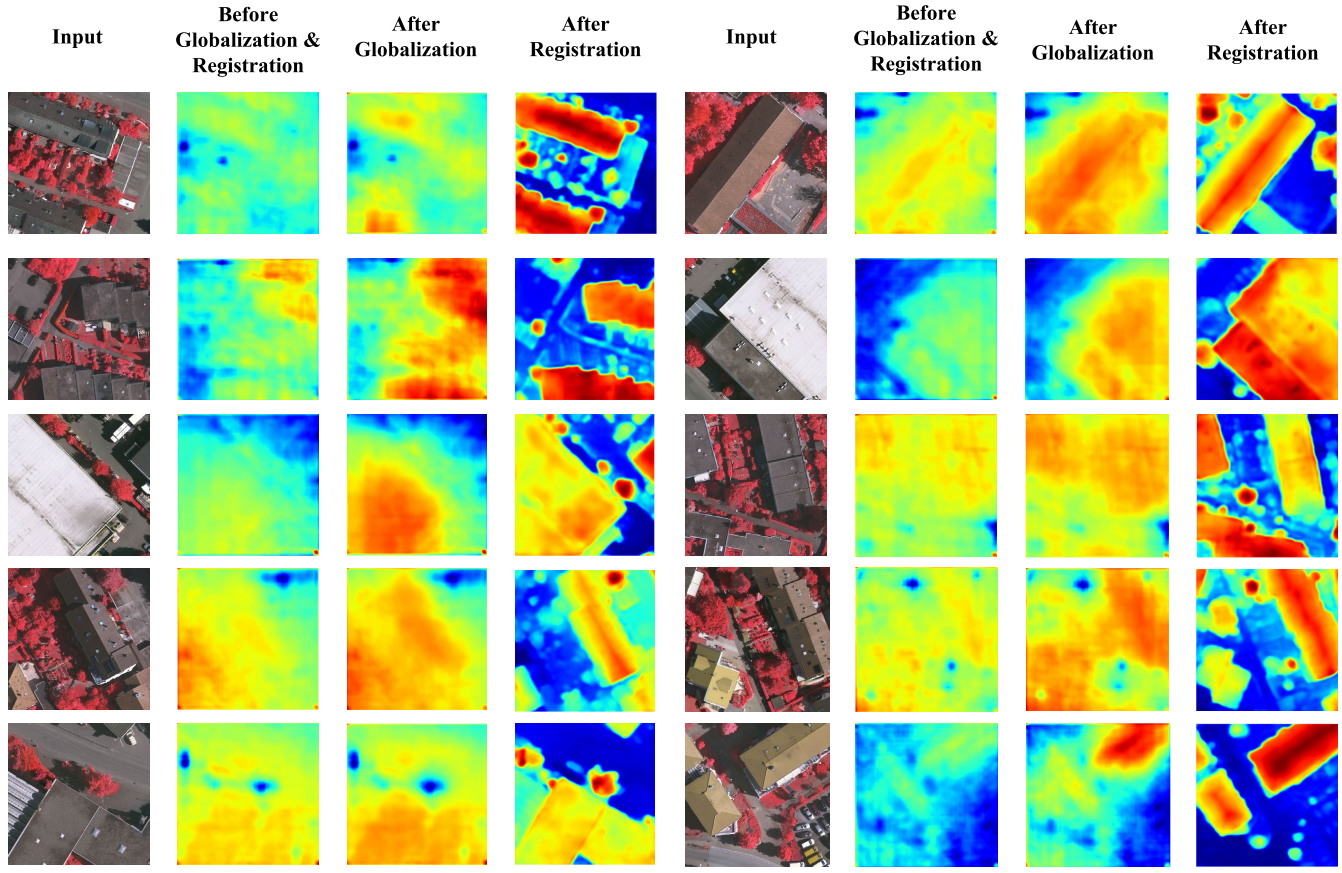


Fig. 9. Visualizations of the feature maps after globalization and registration. The first column is the input of our SFFDE, the second column is the feature before globalization and registration, the third column is the feature map after globalization, and the fourth column is the feature map after registration.

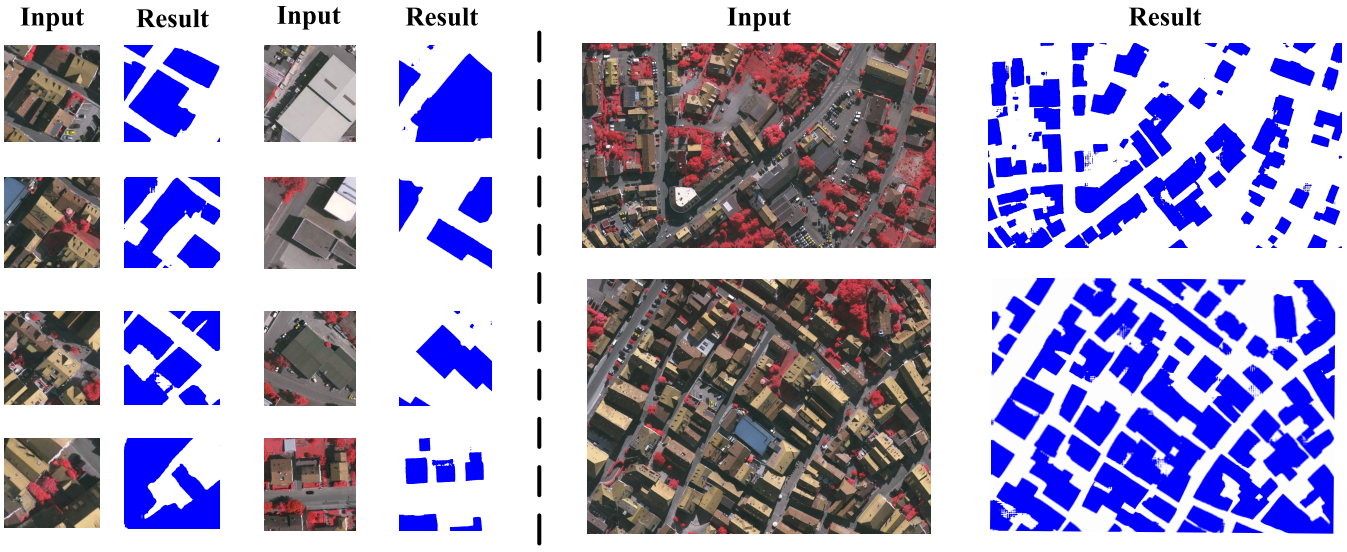


Fig. 10. Visualizations of the building extraction network in our Building3-D framework. (Left) Building extraction results of the image patches. (Right) Extraction results of large area buildings.

the point cloud, which is given in the third column of the figure. Based on the recovered 3-D point cloud, the surface reconstruction results of the building are shown in the fourth column of the figure. It is clear that our framework generates high-quality 3-D point cloud data and builds a 3-D model when there is only a single-view image. This provides an insightful idea for the rapid 3-D reconstruction of large-scale buildings.

3) CityGML Reconstruction: For the reconstruction of the CityGML model of the buildings, our Building3-D adopts the 3dfier [71] approach. The input is the point cloud data obtained in the previous steps and the polygon structure of the

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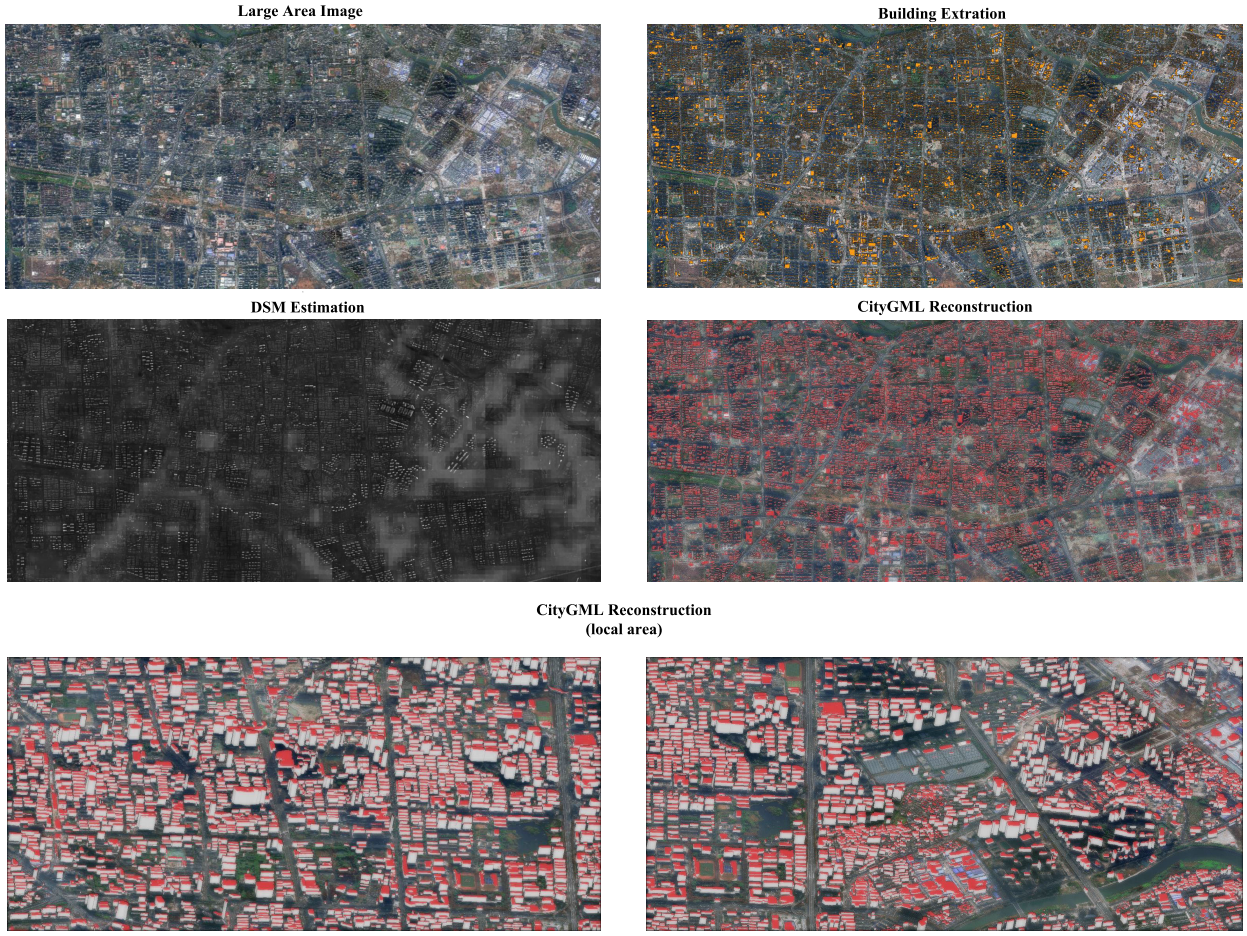


Fig. 11. Visualization results of large area building reconstruction. The large image is obtained from the urban area of Hefei.

buildings, and the output is the LOD1 model of the buildings. The reconstruction results are given in Fig. 8. It is obvious that the building elevation information predicted by our SFFDE can restore the building elevation well, laying a foundation for the 3-D reconstruction of different forms of buildings.

F. Extension

1) Large Area Building Reconstruction: We test the robustness experiments on remote sensing images of the whole urban area in Hefei, Anhui Province, China. As shown in Fig. 11, we give the remote sensing images of the entire area, the building extraction results, the DSM prediction results, and the building LOD1 model reconstruction results, respectively.

G. Ablation Study

We conduct ablation studies on the ISPRS Vaihingen dataset to verify the effectiveness of our SFFDE. PSPNet [59] is selected as the baseline.

1) ESG: As shown in Table III, with ESG, our SFFDE with ResNet101 as the backbone improves the DSM estimation performance by **0.082** (0.276 versus 0.358) on Rel, **0.011** (1.282 versus 1.293) on RMSE, and **0.019** (0.111 versus 0.130) on RMSE(log), which validates that the features with global dependence improve estimation performance more significantly. Before globalization, features of the network are limited to local perception so that the perception of texture regular and irregular instances cannot be provided at the same time. However, after adding ESG, the perception ability of the network is extended from local to global, laying the foundation for subsequent high-level semantic extraction. At the same time, the introduction of ESG also provides a priori global features for the subsequent registration of global and local features.

Channel Number of ESG: In order to better utilize the globalization ability of ESG operations, we conduct ablation experiments on the number of input feature channels. For better visualization, we show spider plots with different channel numbers, as shown in Fig. 12. In order to better display the spider chart, we have performed negative index operations (e^{-x}, where x is the corresponding indicator) on the three indicators of Rel, RMSE, and RMSE(log). Based on this, the six indicators are all closer to the periphery on behalf of higher performance. As can be seen from Fig. 12, when the number of channels is 256, the values of all indicators are located at the outermost periphery, which means that ESG has the best performance at this time.

2) Backbone Selection: To verify the influence of different backbones on our method, we conduct experiments on resnet50 and resnet101, respectively. As shown in the table, we add the proposed ESG operation and L2G-ESR operation

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TABLE IV

ABLATION STUDIES ON AGGREGATION OPERATION. "CONCAT JOINT CONV" DENOTES CONCATENATION JOINT CONVOLUTION. "ELEMENT-WISE ADD" REPRESENTS THE ELEMENT-WISE ADDITION OPERATION. ↑ MEANS THAT THE HIGHER THE INDICATOR VALUE, THE BETTER THE PERFORMANCE, AND ↓ MEANS THAT THE LOWER THE INDICATOR VALUE, THE BETTER THE PERFORMANCE

Aggregation Operation	Rel↓	RMSE↓/m	RMSE(log)↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
Concat joint Conv	0.263	1.252	0.101	0.563	0.718	0.919
Element-wise Add	**0.225**	**1.145**	**0.087**	**0.624**	**0.841**	**0.933**

TABLE V

ABLATION STUDIES ON LOSS FUNCTIONS. ↑ MEANS THAT THE HIGHER THE INDICATOR VALUE, THE BETTER THE PERFORMANCE, AND ↓ MEANS THAT THE LOWER THE INDICATOR VALUE, THE BETTER THE PERFORMANCE

Loss Function	Rel↓	RMSE↓/m	RMSE(log)↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
L1Loss	0.298	1.265	0.100	0.548	0.820	0.912
MSELoss	0.293	1.210	0.099	0.543	0.836	**0.933**
berHuLoss	**0.225**	**1.145**	**0.087**	**0.624**	**0.841**	**0.933**

TABLE VI

COMPARISON OF PERFORMANCE AND INFERENCE TIME. * DENOTES THE NETWORK REPRODUCED BY OURSELVES. CPU: INTEL[1] XEON[1] SILVER 4210R CPU @ 2.40-GHz GPU: NVIDIA RTX 3090

Method	RMSE↓/m	δ_1 ↑	δ_2 ↑	δ_3 ↑	CPU Time(ms)	GPU Time(ms)
Li *et al.** [44]	1.698	0.451	0.817	0.939	8879	115
ASSEH [67]	1.144	0.595	0.860	**0.971**	6538	-
SFFDE + ResNet50 (ours)	1.145	**0.624**	0.841	0.933	5695	85
SFFDE + ResNet101 (ours)	**1.133**	0.595	**0.897**	0.970	8905	118

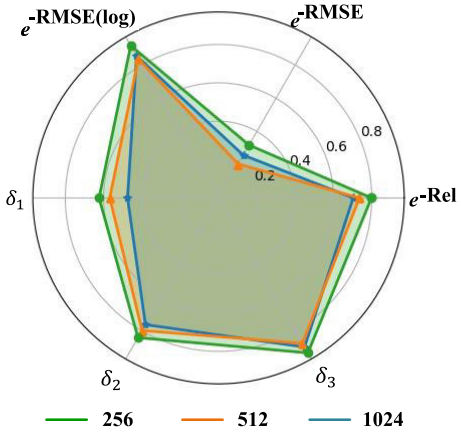


Fig. 12. Ablation studies of the channel number of ESG.

on the basis of resnet50 and resnet101, respectively. We make a horizontal comparison of the backbone, and then, we come to the conclusion that the prediction accuracy of selecting resnet101 as the backbone as a whole is higher than resnet50. Similarly, after adding ESG and L2G-ESR, the overall performance of the resnet101-based network is also better than that of resnet50. This is consistent with our prior knowledge perception.

3) L2G-ESR: In Table III, L2G-ESR further improves the performance by **0.054** (0.222 versus 0.276) on Rel, **0.149** (1.133 versus 1.282) on RMSE, and **0.027** (0.084 versus 0.111) on RMSE(log), which validates that L2G-ESR implements the registration and tradeoff of local and global features. This clearly demonstrates the superiority of SFFDE over other methods of local and global feature representation. After L2G-ESR, the features of the network are locally and globally registered through the concept of a defined ESF. This enables the network to perceive the surrounding pixel features finely in both global and local structures.

Aggregation Operation $\mathcal{I}$: As shown in (13), there are two options in our L2G-ESR feature aggregation: 1) concatenate operation combined with convolution and 2) element-wise addition. We conduct ablation experiments for these two operations, and the experimental results are shown in Table IV. As can be seen from the table, element-wise addition achieves higher performance. Compared with the operation of concatenation joint convolution, element-by-element addition not only does not add redundant parameters but also enables the network to run efficiently.

4) Loss Function: In order to choose the best loss function as the learning direction of the network, we choose L1loss, MSEloss, and berHuLoss for ablation experiments. As shown in Table V, choosing berHuloss as the loss function for network training enables the network to achieve the best prediction performance. Nonetheless, choosing L1loss and MSEloss also achieves impressive performance, in contrast to the baseline. This site reflects the effectiveness of our ESG and L2G-ESR.

5) Performance Spider Plots: To illustrate the predicted performance of our SFFDE, we visualize the spider plots of the performance (see Fig. 13). We compare with Li et al. [44] and Amirkolaee and Arefi [41] on the ISPRS dataset. On the

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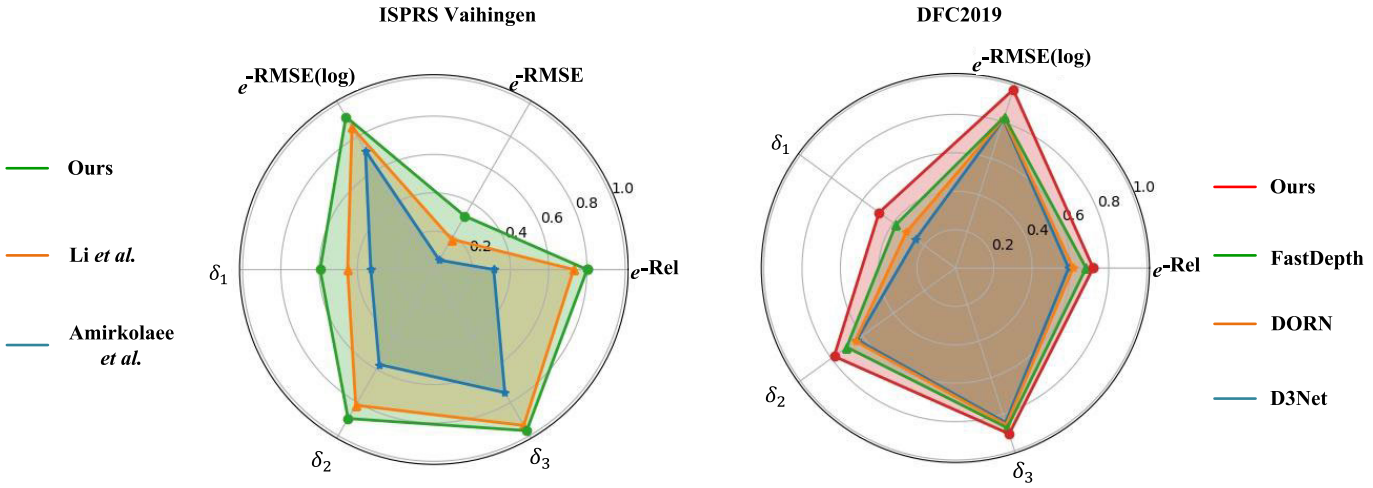


Fig. 13. Performance spider plots on the ISPRS Vaihingen dataset and the DFC2019 dataset.

DFC2019 dataset, comparisons with D3Net [45], DORN [73], and FastDepth [74] are given. Since the three indicators of Rel, RMSE, and RMSE(log) are lower, the better, so we carry out negative index processing (e^{-x}). Based on this, these six indicators are all as high as possible. We present the performance spider plots for the ISPRS Vaihingen dataset and the DFC2019 dataset, respectively. Since the higher the indicator value, the better the performance, the closer the indicator value of the spider plot is to the periphery, and the higher the performance. It is obvious that our SFFDE is at the outermost periphery of the spider plot, whether it is the ISPRS Vaihingen dataset or the DFC2019 dataset. Therefore, this shows that our SFFDE outperforms existing methods in either metric.

6) Inference Time Analysis: As shown in Table VI, we present the performance and inference time comparison of our method with existing methods. When using Resnet50 as the backbone, our method outperforms Li et al. by a large margin and has a shorter inference time. Furthermore, our method is comparable in performance to ASSEH, but our method has a shorter inference time. When using Resnet101 as the backbone, our method still outperforms Li et al. by a large margin and is comparable in inference time. Furthermore, our method outperforms ASSEH in performance but slightly prolongs inference time. In general, our method has superior inference time while ensuring high performance, which also effectively verifies the superiority of our method.

V. Conclusion

We propose a framework (Building3-D) for creating 3-D building models from single-view remote sensing imagery. Our Building3-D is rooted in the proposed SFFDE network to achieve globalization of semantics and registration of global features with local features through the proposed ESG and L2G-ESR. Extensive experiments on the commonly used ISPRS Vaihingen and DFC2019 datasets demonstrate the superiority of SFFDE for DSM estimation, providing accurate elevation information for building reconstruction, and δ_1, δ_2, and δ_3 metrics of our SFFDE are improved to 0.595, 0.897, and 0.970. Building3-D achieves 3-D reconstruction of large-area buildings in a single-view image by utilizing SFFDE elevation estimation, building mask extraction, point cloud reconstruction, and building reconstruction, which is in sharp contrast to other methods based on multiview images. Building3-D not only reduces data acquisition costs but also enables rapid and large-area building reconstruction. As a novel-and-efficient method, Building3-D provides a fresh perspective into the challenging 3-D reconstruction of buildings. Although the 3-D reconstruction effect is still greatly improved, we will continue to study to generate more refined 3-D reconstruction results.

For the lattice-like stripes phenomenon, we will work on future research. First, we will focus on large-scale images as input to build a neural network for learning, which greatly reduces the number of patch edges. In addition, we will also work on the design of more accurate interpatch fusion methods. For the overall network architecture, although our Building3-D adopts the method of elevation estimation and building extraction parallel reasoning in the overall architecture to improve efficiency, the number of network parameters has increased. In view of this, we plan to introduce a multitask learning method in the follow-up research, by inputting a single image block and simultaneously performing multitask output (elevation regression head and building extraction head).

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